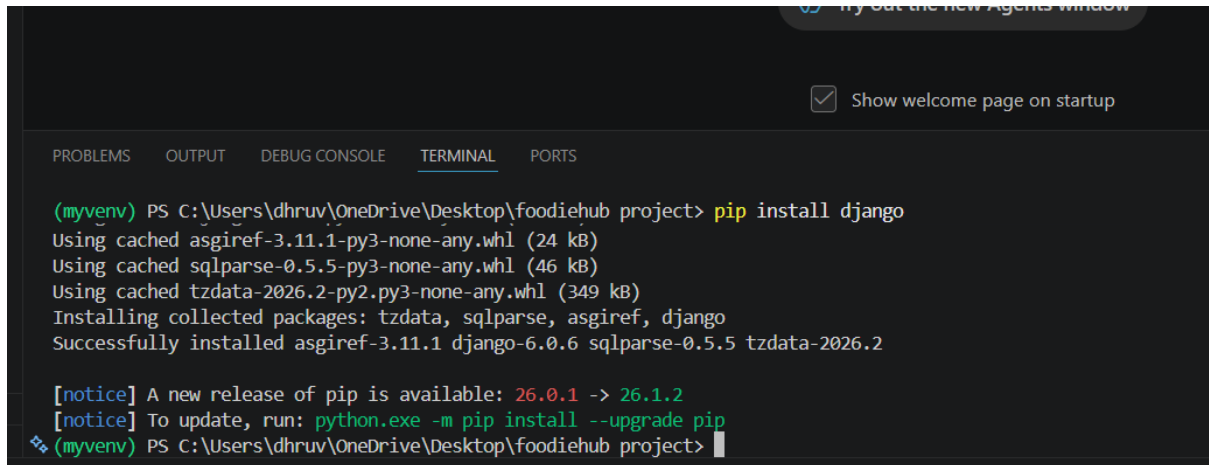


1. Install Django using pip in your local Python environment and verify the installation by running 'django-admin --version' in the terminal.

Ans=



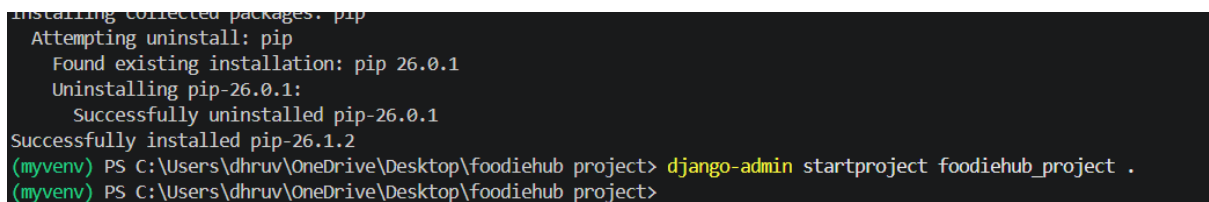
The screenshot shows a VS Code terminal window with the following content:

```
(myvenv) PS C:\Users\dhruv\OneDrive\Desktop\foodiehub project> pip install django
Using cached asgiref-3.11.1-py3-none-any.whl (24 kB)
Using cached sqlparse-0.5.5-py3-none-any.whl (46 kB)
Using cached tzdata-2026.2-py2.py3-none-any.whl (349 kB)
Installing collected packages: tzdata, sqlparse, asgiref, django
Successfully installed asgiref-3.11.1 django-6.0.6 sqlparse-0.5.5 tzdata-2026.2

[notice] A new release of pip is available: 26.0.1 -> 26.1.2
[notice] To update, run: python.exe -m pip install --upgrade pip
(myvenv) PS C:\Users\dhruv\OneDrive\Desktop\foodiehub project>
```

2. Create a new Django project named 'foodiehub_project' using the django-admin startproject command, then run the development server and open <http://127.0.0.1:8000/> to confirm it's working.

Ans=



The screenshot shows a VS Code terminal window with the following content:

```
Installing collected packages: pip
Attempting uninstall: pip
Found existing installation: pip 26.0.1
Uninstalling pip-26.0.1:
Successfully uninstalled pip-26.0.1
Successfully installed pip-26.1.2
(myvenv) PS C:\Users\dhruv\OneDrive\Desktop\foodiehub project> django-admin startproject foodiehub_project .
(myvenv) PS C:\Users\dhruv\OneDrive\Desktop\foodiehub project>
```

3. Explore the folder structure of your 'foodiehub_project' and write down the purpose of each key file: manage.py, settings.py, urls.py, wsgi.py, and asgi.py.

Hint: Check the official Django documentation or use ChatGPT to help you describe each file's role.

Ans=

File Name Purpose

manage.py is a command-line utility used to manage the Django project. It helps run the development server, create migrations, create apps, and perform other administrative tasks.

settings.py contains all the project settings such as installed apps, database configuration, templates, static files, security settings, and other project configurations.

urls.py defines the URL patterns for the project. It connects web addresses (URLs) to their corresponding view functions or classes.

wsgi.py is used for deploying the Django project on WSGI-compatible web servers such as Gunicorn or Apache with mod_wsgi. It acts as the entry point for handling web requests in synchronous applications.

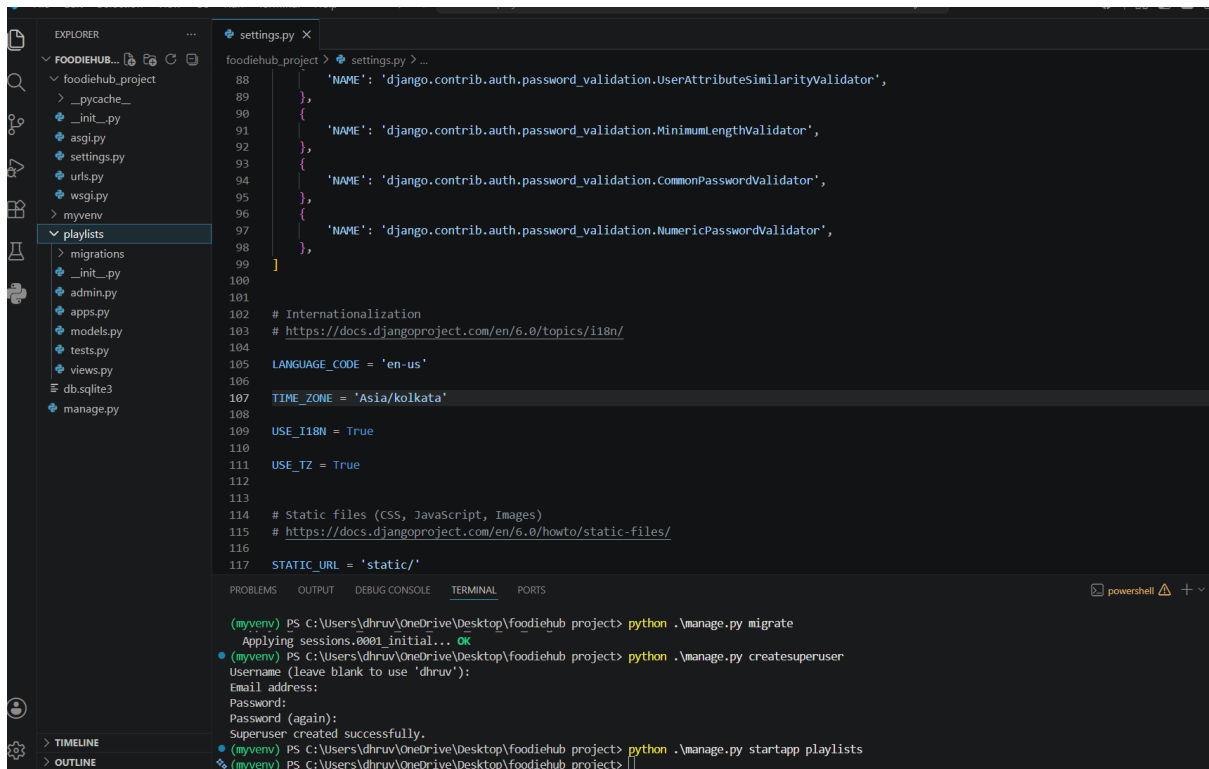
asgi.py is used for deploying Django on ASGI-compatible servers such as Uvicorn or Daphne. It supports asynchronous features like WebSockets, real-time applications, and async views.

4. Write a short comparison (3-4 lines) between Django and Flask, focusing on features and use cases for each. Use an example app you use daily (like Instagram or Zomato) to explain which framework would be better and why.

Ans=

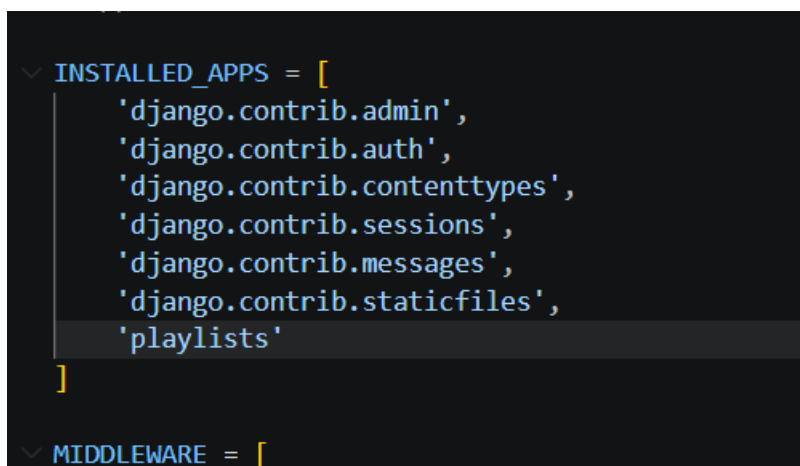
Django is a full-featured Python web framework that includes built-in features like authentication, an admin panel, and database management, making it ideal for large and complex web applications. Flask is a lightweight framework that gives developers more flexibility and is better suited for small projects and APIs. For an app like Instagram, Django is the better choice because it needs user authentication, database support, security, and scalability to handle millions of users efficiently.

5. Create a new Django app named 'playlists' inside your existing Django project and confirm that a playlists folder with the standard files (views.py, models.py, etc.) is generated.



6. Add 'playlists' to the INSTALLED_APPS list in your project's settings.py file so that Django recognizes it.

Ans =



7. In playlists/views.py, write a function home(request) that returns an HttpResponse displaying 'Welcome to My Spotify Playlists!'.

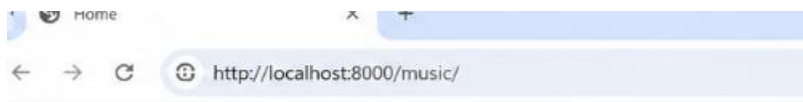
Ans=

Welcome to My Spotify Playlists

This is the homepage for your playlists app.

8. Configure your project's `urls.py` to map the URL path `'music/'` to the home view you created in `playlists/views.py`, so visiting `http://localhost:8000/music/` shows your welcome message.

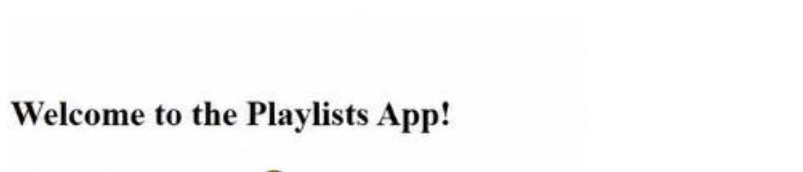
Ans=



9. Change the welcome message in your `playlists/views.py` home view to include your name, then restart the server and verify the update appears at `http://localhost:8000/music/`.

Hint: This helps you see how Django reloads views and how URL mapping works in real time.

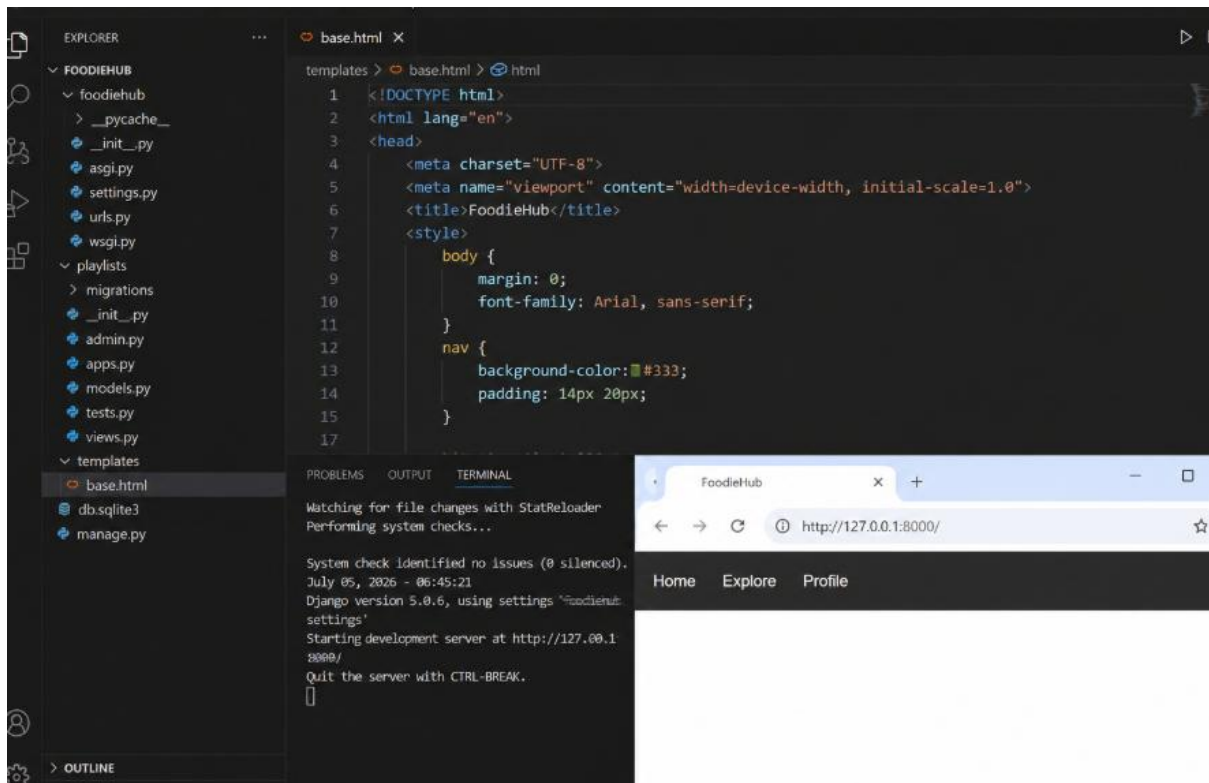
Ans=



Hello, I am **Dhruv**!

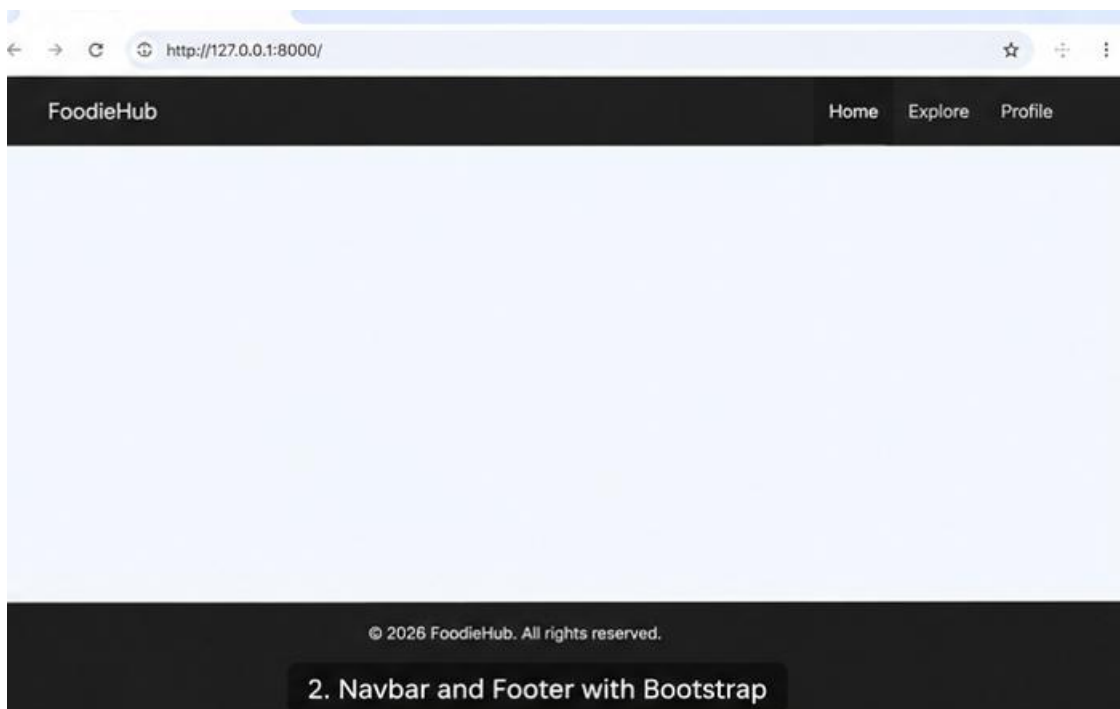
10. Create a `templates` folder in your Django or Flask project, and add a new file called `base.html` that includes a simple navigation bar with links for 'Home', 'Explore', and 'Profile'—use HTML only for now.

Ans=



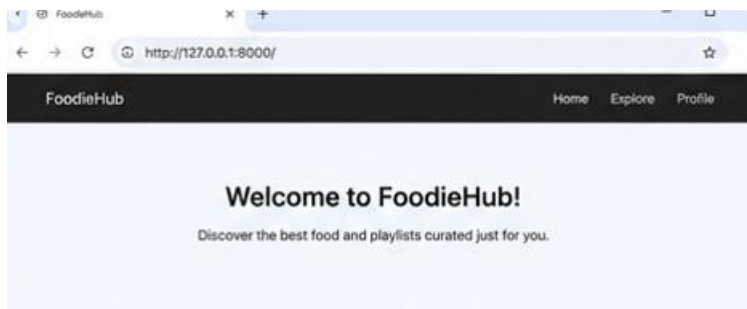
11. Use Bootstrap via CDN in your base.html to style the navigation bar and add a footer section that remains at the bottom of the page.

Hint: Copy the Bootstrap CDN link from <https://getbootstrap.com> and use Bootstrap classes like navbar, container, and footer.



12. Inside your templates folder, create a new home.html file that extends base.html using template inheritance, and add a welcome message in the main content area.

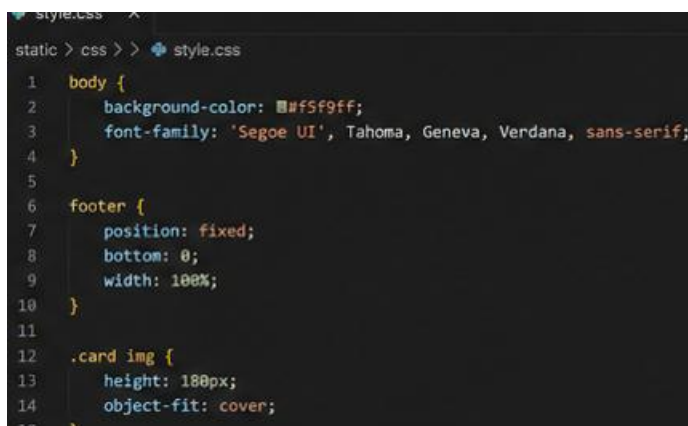
Ans=



13. Add a static folder to your project and place a custom CSS file (style.css) inside it. Link this static CSS file in base.html using the correct template tag, and change the background color of the body.

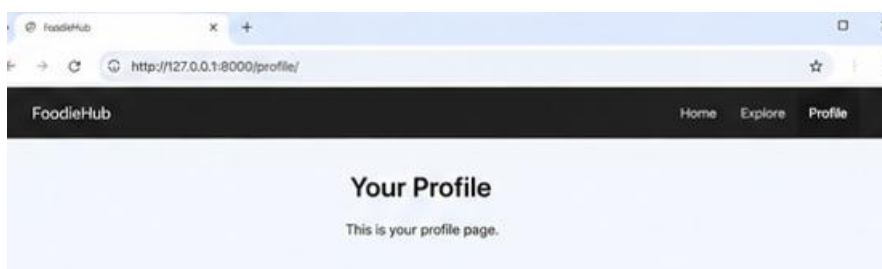
Hint: Use the {% load static %} tag and the {% static 'style.css' %} syntax for linking.

Ans=

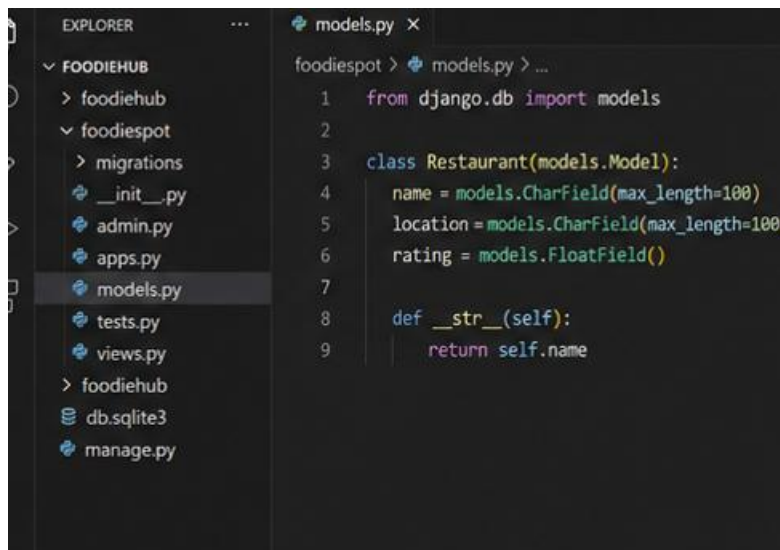


14. Create a new explore.html template that extends base.html and displays a grid of 6 cards (use Bootstrap card component) with dummy images and titles—imagine these as trending playlists like on Spotify or trending restaurants like on Zomato.

Ans=



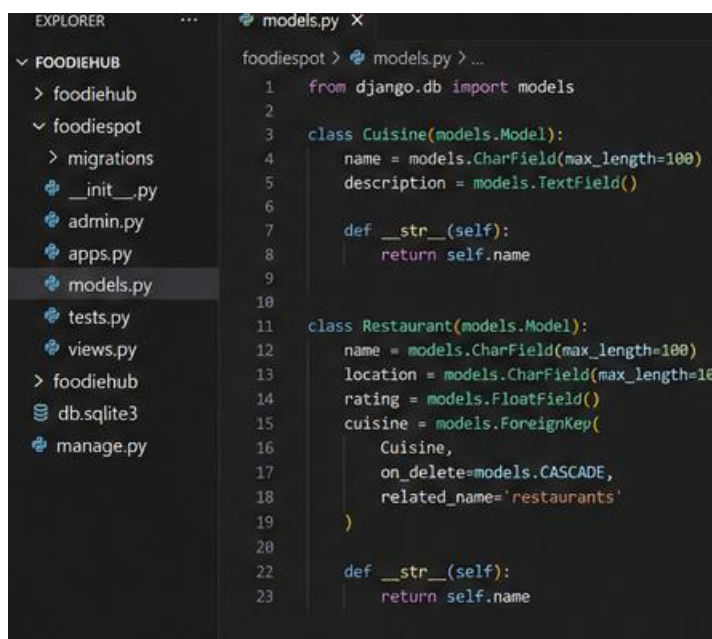
15. Create a Django model named Restaurant with fields name (CharField), location (CharField), and rating (FloatField) in a new app called foodiespot.



The screenshot shows the Django IDE with the Explorer on the left and the code editor on the right. The Explorer shows the project structure with 'FOODIEHUB' and 'foodiespot' apps. The 'foodiespot' app is expanded, showing files like 'migrations', 'models.py', 'tests.py', and 'views.py'. The 'models.py' file is selected. The code editor shows the following Python code:

```
foodiespot > models.py > ...
1  from django.db import models
2
3  class Restaurant(models.Model):
4      name = models.CharField(max_length=100)
5      location = models.CharField(max_length=100)
6      rating = models.FloatField()
7
8      def __str__(self):
9          return self.name
```

16. Add a Cuisine model to your foodiespot app with fields: name (CharField) and description (TextField), then set up a ForeignKey in Restaurant to link each restaurant to a cuisine.



The screenshot shows the Django IDE with the Explorer on the left and the code editor on the right. The Explorer shows the project structure with 'FOODIEHUB' and 'foodiespot' apps. The 'foodiespot' app is expanded, showing files like 'migrations', 'models.py', 'tests.py', and 'views.py'. The 'models.py' file is selected. The code editor shows the following Python code:

```
EXPLORER    ...    models.py X
FOODIEHUB
> foodiehub
> foodiespot
  > migrations
  > __init__.py
  > admin.py
  > apps.py
  > models.py
  > tests.py
  > views.py
> foodiehub
db.sqlite3
manage.py

foodiespot > models.py > ...
1  from django.db import models
2
3  class Cuisine(models.Model):
4      name = models.CharField(max_length=100)
5      description = models.TextField()
6
7      def __str__(self):
8          return self.name
9
10
11 class Restaurant(models.Model):
12     name = models.CharField(max_length=100)
13     location = models.CharField(max_length=100)
14     rating = models.FloatField()
15     cuisine = models.ForeignKey(
16         Cuisine,
17         on_delete=models.CASCADE,
18         related_name='restaurants'
19     )
20
21     def __str__(self):
22         return self.name
23
```

17. Run makemigrations and migrate commands to create the database tables for your Restaurant and Cuisine models, then verify the tables are created in your SQLite database using DB Browser for SQLite or the Django shell.

Ans =

```

PS C:\Users\dhruv\Desktop\foodiehub> python manage.py makemigrations foodiespot
Migrations for 'foodiespot':
  foodiespot\migrations\0001_initial.py
    - Create model Cuisine
    - Create model Restaurant
PS C:\Users\dhruv\Desktop\foodiehub> python manage.py migrate
Operations to perform:
  Apply all migrations: admin, auth, contenttypes, foodiespot, sessions
Running migrations:
  Applying foodiespot.0001_initial... OK

```

18. Change your Django project's database settings to connect to a MySQL database using pymysql, and update your foodiespot app's models to work with MySQL instead of SQLite.

Hint: Install pymysql and update DATABASES in settings.py. Make sure MySQL server is running and you have created the database.

Ans=

```

# settings.py X
foodiehub > settings.py > DATABASES
1  # MySQL Database Configuration
2  import pymysql
3  pymysql.install_as_MySQLdb()
4
5  DATABASES = {
6      'default': {
7          'ENGINE': 'django.db.backends.mysql',
8          'NAME': 'foodiehubdb',
9          'USER': 'root',
10         'PASSWORD': 'root123',
11         'HOST': '127.0.0.1',
12         'PORT': '3306',
13         'OPTIONS': {
14             'charset': 'utf8mb4',
15         }
16     }
17 }

```

19. Use Django's ORM to add three new Restaurant objects and two Cuisine objects, then write a query to get all restaurants with a rating above 4.0 and print their names.

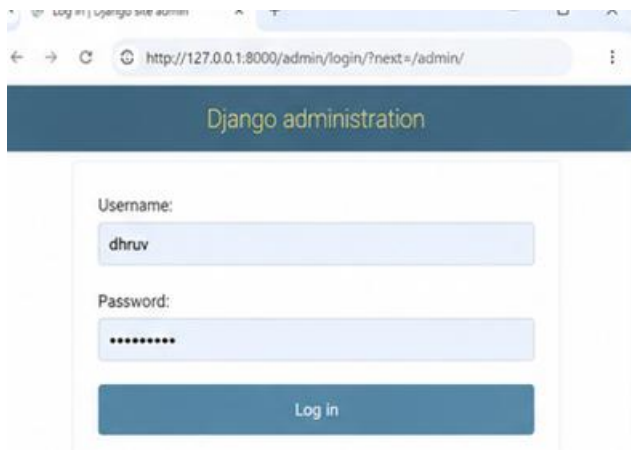

```
>>> from foodiespot.models import Cuisine, Restaurant
>>> italian = Cuisine.objects.create(name='Italian', description='Classic Italian dishes')
>>> indian = Cuisine.objects.create(name='Indian', description='Spicy and flavorful Indian food')

>>> Restaurant.objects.create(name='Pasta Palace', location='New York', rating=4.5, cuisine=italian)
<Restaurant: Pasta Palace>
>>> Restaurant.objects.create(name='Spice Hub', location='Delhi', rating=4.7, cuisine=indian)
<Restaurant: Spice Hub>
>>> Restaurant.objects.create(name='Pizza Point', location='Chicago', rating=3.9, cuisine=italian)
<Restaurant: Pizza Point>

# Get all restaurants with rating > 4.0
>>> high_rated = Restaurant.objects.filter(rating__gt=4.0)
>>> for r in high_rated:
...     print(r.name)
```

20.

Enable the Django admin interface in your project and create a superuser account so you can log in to the admin dashboard.



21. Create a new model called Restaurant with fields name, cuisine, and rating, then register it with the Django admin so it appears in the admin panel.

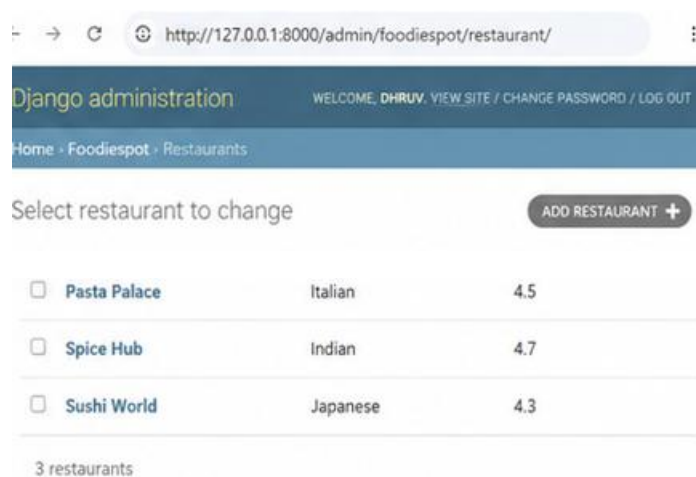
Ans=

```
1 from django.db import models
2
3 class Restaurant(models.Model):
4     name = models.CharField(max_length=100)
4     cuisine = models.CharField(max_length=100)
5     rating = models.FloatField()
6
7     def __str__(self):
18         return self.name
```

```
admin.py x
1 from django.contrib import admin
2 from .models import Restaurant
3
4 # Register your models here.
5 admin.site.register(Restaurant)
```

22. Customize the Restaurant admin by displaying name, cuisine, and rating in the list view using the list_display option in admin.py.

Ans=



23. Add search functionality in the Restaurant admin so you can search restaurants by name or cuisine using the search_fields option.

Ans=



24. Add a filter in the Restaurant admin panel to filter restaurants by cuisine using the list_filter option. Hint: Use list_filter = ['cuisine'] in your RestaurantAdmin class.

Ans=

Django administration

Home > Foodiespot > Restaurants

Select restaurant to change

Action: 0 of 2 selected

☐	NAME	CUISINE	RATING
☐	Spice Hub	Indian	4.7
☐	Curry Corner	Indian	4.2

2 restaurants (filtered from 4 total)

FILTER

By cuisine

- All
- Indian
- Italian
- Japanese
- Chinese

25. Create a Django Form class named AddRestaurantForm with fields for restaurant name, cuisine type, and contact email, then render it in a template using the GET method.

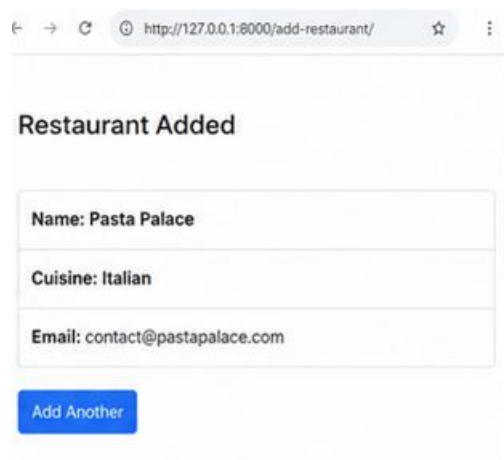
Ans=

```

forms.py | views.py | add_restaurant.html
foodiespot > forms.py
1 from django import forms
2
3 class AddRestaurantForm(forms.Form):
4     name = forms.CharField(
5         label='Restaurant Name',
6         min_length=3,
7         widget=forms.TextInput(attrs={'class':
8                                     'form-control'})
9     )
10    cuisine = forms.ChoiceField(
11        label='Cuisine Type',
12        choices=[
13            ('indian', 'Indian'),
14            ('italian', 'Italian'),
15            ('chinese', 'Chinese'),
16            ('japanese', 'Japanese'),
17        ],
18        widget=forms.Select(attrs={'class':
19                                'form-control'})
20    )
21    email = forms.EmailField(
22        label='Contact Email',
23        widget=forms.EmailInput(attrs=
24                                {'class': 'form-control'})
25    )
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```

26. Modify your AddRestaurantForm to handle form submission using the POST method and display the submitted data on a new page after the user submits the form.

Ans=



Restaurant Added

Name: Pasta Palace

Cuisine: Italian

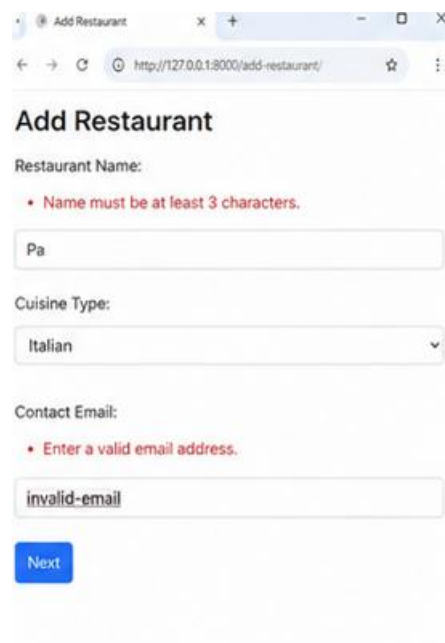
Email: contact@pastapalace.com

Add Another

27. Add validation to your AddRestaurantForm so that the restaurant name must be at least 3 characters and the contact email must be a valid email address. Display any validation errors directly above the form fields in your template.

Hint: Use Django's built-in validators and `{{ form.errors }}` in your template.

Ans=



Add Restaurant

Restaurant Name:

- Name must be at least 3 characters.

Pa

Cuisine Type:

Italian

Contact Email:

- Enter a valid email address.

invalid-email

Next

28. Implement CSRF protection in your form template and verify that submitting the form without the CSRF token results in a forbidden error.

Hint: Use `{% csrf_token %}` inside your form tags.

Ans=

```
</head>
<body class="container mt-4">
  <h2>Add Restaurant</h2>
  <form method="post">
    {X csrf_token X} <!-- CSRF token added -->
    {{ form.as_p }}
    <button type="submit" class="btn btn-primary">Submit</button>
  </form>
</body>
```

29.

Create a Django model called Restaurant with fields name (CharField), cuisine (CharField), and rating (FloatField), then generate a ModelForm named RestaurantForm for it.

Ans=

```
from django.db import models

class Restaurant(models.Model):
    name = models.CharField(max_length=100)
    cuisine = models.CharField(max_length=100)
    rating = models.FloatField()

    def __str__(self):
        return self.name
```

```
from django import forms
from .models import Restaurant

class RestaurantForm(forms.ModelForm):
    class Meta:
        model = Restaurant
        fields = ['name', 'cuisine', 'rating']
```

30. Use RestaurantForm to build a simple Django view that displays the form and saves new restaurant entries to the database when submitted.

Ans=

```

views.py x
restaurantapp > views.py > ...

from django.shortcuts import render, redirect
from .forms import RestaurantForm

def add_restaurant(request):
    if request.method == 'POST':
        form = RestaurantForm(request.POST)
        if form.is_valid():
            form.save()
            return redirect('success')
        else:
            form = RestaurantForm()
            return render(request, 'add_restaurant.html',
                          {'form': form})

def success(request):
    return render(request, 'success.html')

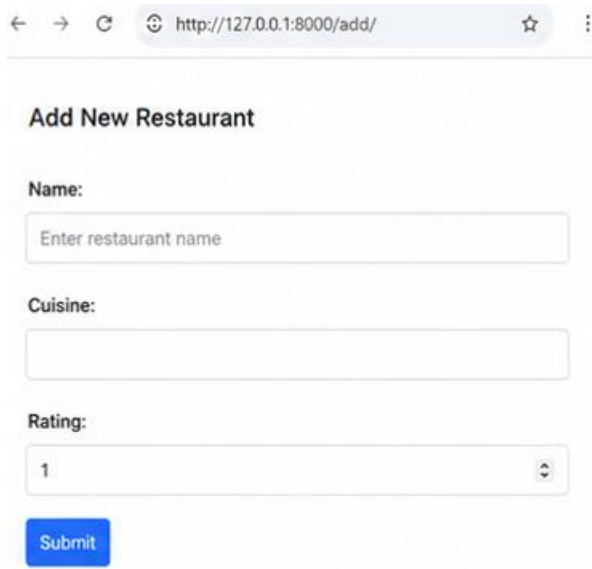
```

The screenshot shows a web browser at the URL `http://127.0.0.1:8000/add/`. The page has a title "Add New Restaurant". It contains three input fields: "Name:" with the value "Pasta Palace", "Cuisine:" with the value "Italian", and "Rating:" with the value "4.5". Below these fields is a blue "Submit" button. A second browser window is shown below, with the URL `http://127.0.0.1:8000/success/`, displaying a green message: "Restaurant added successfully!"

31. Customize the RestaurantForm to add a placeholder for the name field and set the minimum value for the rating field to 1 and maximum to 5.

Hint: Use the widgets attribute inside the ModelForm's Meta class.

Ans=



← → ↻ http://127.0.0.1:8000/add/ ☆ ⋮

Add New Restaurant

Name:

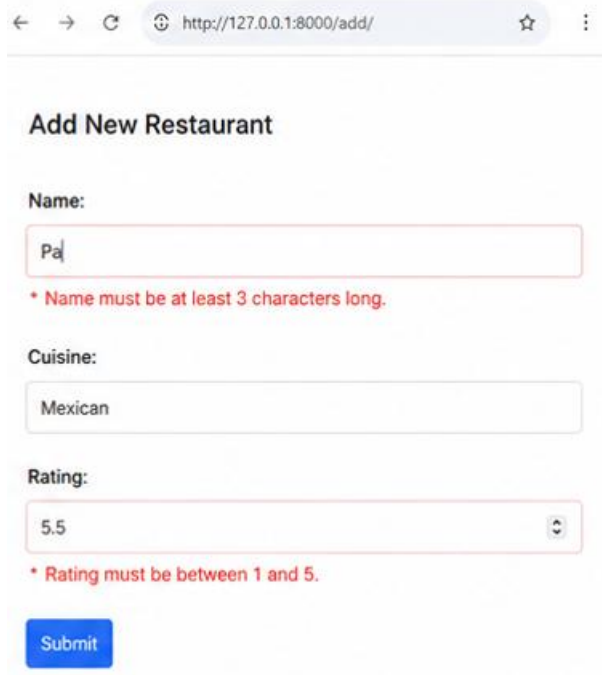
Cuisine:

Rating:

Submit

32. Add validation to RestaurantForm so that the name must be at least 3 characters and the rating must be between 1 and 5. Show appropriate error messages on the form.

Ans=



← → ↻ http://127.0.0.1:8000/add/ ☆ ⋮

Add New Restaurant

Name:

* Name must be at least 3 characters long.

Cuisine:

Rating:

* Rating must be between 1 and 5.

Submit

33. Create a Python Django project called musiczone and add a User model if not already present. Set up the default authentication system so users can register, log in, and log out through Django's built-in views.

Ans=

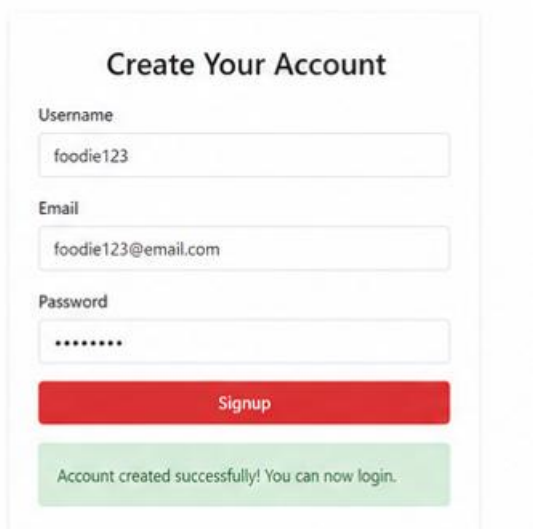
```

musiczone
1  INSTALLED_APPS = [
2      'django.contrib.admin',
3      'django.contrib.auth',
4      'django.contrib.contenttypes',
5      'django.contrib.sessions',
6      'django.contrib.messages',
7      'django.contrib.staticfiles',
8      'accounts',
9  ]
10
11  AUTH_USER_MODEL = 'auth.User'
12
13  LOGIN_URL = 'login'
14  LOGOUT_REDIRECT_URL = 'home'
15  LOGIN_REDIRECT_URL = 'welcome'
16
17  TEMPLATES = [
18      {
19          'BACKEND': 'django.template.backends.django.DjangoTemplates',
20          'DIRS': [BASE_DIR / 'templates'],
21          'APP_DIRS': True,
22          'OPTIONS': {
23              'context_processors': [
24                  'django.template.context_processors.debug',
25                  'django.template.context_processors.request',
26                  'django.contrib.auth.context_processors.auth',
27              ],
28          },
29      },
30  ]

```

34. Build a signup form for a food delivery app (like Zomato) using Django. The form should collect username, email, and password, and save new users securely with hashed passwords.

Ans=



35. Implement a login feature for a movie booking site (BookMyShow style) using Django's authentication backend. After login, redirect the user to a welcome page showing their username.

Login

Username

moviebuff

Password

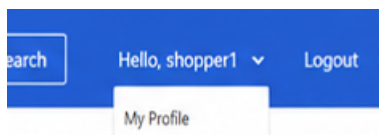
Login

Don't have an account? [Signup](#)

36. Display the logged-in user's username in the navbar of your e-commerce app (Flipkart-style) template. Show 'Login' and 'Signup' links if no user is logged in, and 'Logout' if a user is logged in.

Hint: Use Django's template tags like `{% if user.is_authenticated %}` to conditionally render navbar items.

Ans=



37. Configure SMTP email settings in your Django project's settings.py file to use Gmail for sending emails. Test your setup by sending a simple test email using Django's `send_mail` function.

Ans=

```

settings.py x
myproject > settings.py > ...
188 # Email backend settings (Gmail SMTP)
189 EMAIL_BACKEND = 'django.core.mail.backends.smtp.EmailBackend'
190 EMAIL_HOST = 'smtp.gmail.com'
191 EMAIL_PORT = 587
192 EMAIL_USE_TLS = True
193 EMAIL_HOST_USER = 'yourgmail@gmail.com'
194 EMAIL_HOST_PASSWORD = 'your-app-password' # Use App Password
195 DEFAULT_FROM_EMAIL = EMAIL_HOST_USER

```

38. Create a Django view function `send_password_reset_email(request, user_email)` that sends a password reset link to the given email address using `send_mail`. Use a subject line and plain text message body similar to what you see in apps like Paytm or Flipkart.

Ans=

```

from django.core.mail import send_mail
from django.shortcuts import render
from django.conf import settings

def send_password_reset_email(request, user_email):
    subject = 'Reset Your Password - Paytm'
    reset_link = f'http://127.0.0.1:8000/reset-password/{user_email}/'
    message = f"""
    Hi,
    We received a request to reset your password.
    Click the link below to set a new password:
    {reset_link}
    If you did not request this, please ignore this email.
    Thanks,
    Paytm Team
    """
    send_mail(
        subject,
        message,
        settings.DEFAULT_FROM_EMAIL,
        [user_email],
        fail_silently=False,
    )
    return render(request, 'email_sent.html', {'email': user_email})

```

39. Design an HTML email template called `order_confirmation.html` for a Swiggy-style food order confirmation. Use Django's `EmailMultiAlternatives` to send an email with this HTML template when a user places an order.

Hint: Use Django's `render_to_string` to generate the HTML content from your template.

Ans=

```

<!DOCTYPE html>
<html>
<body style="font-family: Arial, sans-serif; background:#f5f5f5;">
<div style="max-width:600px;margin:0 auto;background:#fff;padding:20px;">
<div style="text-align:center;">

</div>
<h2 style="color:#fc8d62;">Order Confirmed! 🍕</h2>
<p>Thank you for ordering with <b>Swiggy</b>. Your order is confirmed.</p>
<hr>
<h3>Order Details</h3>
<ol>
<li><b>Order ID:</b> {{ order.id }}</li>
<li><b>Customer:</b> {{ user_name }}</li>
<li><b>Items:</b> {{ items }}
<ul>
<li>{{ for item in items }}
<li>{{ item.name }} - {{ item.qty }} x ₹{{ item.price }}</li>
<li>{{ endfor }}</li>
</ul>
<li><b>Total:</b> ₹{{ total }}</li>
</ol>
<p style="margin-top:20px;">We will deliver your food soon! 🍕</p>
<p>- Swiggy Team</p>
</body>
</html>

```

40. Modify your HTML email template to include the user's name and order details dynamically using Django template variables. Test by sending a sample order confirmation email to yourself.

```

from django.core.mail import EmailMultiAlternatives
from django.template.loader import render_to_string

def send_order_confirmation(user_email, user_name, order):
    html_content = render_to_string('emails/order_confirmation.html', {
        'user_name': user_name,
        'order': order,
        'items': order.items.all(),
        'total': order.get_total,
    })
    subject = f"Order #{order.id} Confirmed - Swiggy"
    text_content = f"""
    Hi {user_name},
    Your order #{order.id} has been confirmed.
    Total: ₹{order.get_total}

    Thanks,
    Swiggy Team
    """

    msg = EmailMultiAlternatives(subject, text_content,
                                settings.DEFAULT_FROM_EMAIL, [user_email])
    msg.attach_alternative(html_content, "text/html")
    msg.send()
    return True

```

41.

Use ChatGPT or Copilot to help you generate a catchy subject line and body for a 'Welcome to IPL Fantasy League' HTML email. Implement the email sending logic in a Django view and send the email to a test user.

Hint: Paste your AI-generated subject and body into your code, then send the email using EmailMultiAlternatives.

Ans=

```

from django.core.mail import EmailMultiAlternatives
from django.template.loader import render_to_string

def send_ipl_welcome_email(user_email, user_name):
    html_content = render_to_string('emails/ipl_welcome.html', {
        'user_name': user_name,
    })
    subject = "🔥 Welcome to IPL Fantasy League - Let the Games Begin!"

    text_content = f"""
    Hi {user_name},
    Welcome to the IPL Fantasy League!
    Get ready to build your dream team, score big
    and win exciting prizes.
    Let the battle for glory begin!

    Play now: http://127.0.0.1:8000/ipl/

    Cheers,
    IPL Fantasy League Team
    """

    msg = EmailMultiAlternatives(subject, text_content,
                                settings.DEFAULT_FROM_EMAIL, [user_email])
    msg.attach_alternative(html_content, "text/html")
    msg.send()
    return True

```

42. Create a Django view called forgot_password_view that renders a form where users enter their email to request a password reset.

```

from django.shortcuts import render, redirect
from django.core.mail import send_mail
from django.conf import settings
import random

def forgot_password_view(request):
    if request.method == 'POST':
        email = request.POST.get('email')
        if email:
            otp = str(random.randint(100000, 999999))
            request.session['otp'] = otp
            request.session.set_expiry(300) # 5 minutes

            subject = 'Your Password Reset OTP'
            message = f'Your OTP for password reset is {otp}. It is valid for 5 minutes.'
            ..
            send_mail(
                subject,
                message,
                settings.DEFAULT_FROM_EMAIL,
                [email],
                fail_silently=False,
            )
            return redirect('verify_otp')

    return render(request, 'forgot_password.html')

```

Forgot Password

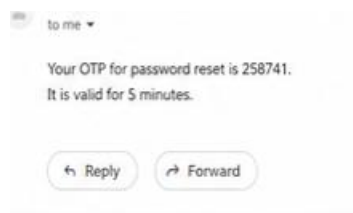
Enter your email address

Send OTP

43. Generate a random 6-digit OTP in your Django view, store it in the session, and send it to the user's email using Django's send_mail function.

Hint: Use request.session['otp'] to save the OTP and settings.py to configure email backend for testing.

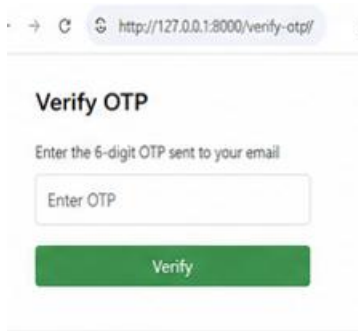
Ans=



44. Build a Django form where the user enters the OTP they received via email; validate the OTP by comparing it with the value stored in the session, and display a success or error message accordingly.

Ans=

```
def verify_otp_view(request):
    if request.method == 'POST':
        user_otp = request.POST.get('otp')
        session_otp = request.session.get('otp')
        if user_otp and session_otp:
            if user_otp == session_otp:
                del request.session['otp']
                return render(request,
                              'otp_success.html')
            else:
                return render(request,
                              'otp_invalid.html')
        return render(request, 'verify_otp.html')
```



45. Set a session expiry time of 5 minutes for the OTP in your Django app so that the OTP becomes invalid after that period.

Hint: Use `request.session.set_expiry(300)` after setting the OTP.

```
def forgot_password_view(request):
    if request.method == 'POST':
        email = request.POST.get('email')
        if email:
            otp = str(random.randint(100000, 999999))
            request.session['otp'] = otp
```

50. Create a custom Django middleware called `BlockExpiredOTPAccessMiddleware` that checks if the OTP session key is missing or expired, and if so, redirects users trying to access the OTP verification page back to the forgot password form.

```
from django.shortcuts import redirect

class BlockExpiredOTPAccessMiddleware:
    def __init__(self, get_response):
        self.get_response = get_response

    def __call__(self, request):
        otp_paths = ['/verify-otp/', '/otp-success/',
                     '/otp-invalid/']
        if request.path in otp_paths:
            otp = request.session.get('otp')
            if not otp:
                return redirect("forgot_password")

        response = self.get_response(request)
        return response
```

51. Set up MEDIA_ROOT and MEDIA_URL in your Django project's settings.py so that uploaded media files are stored in a folder named 'user_uploads' and can be accessed via '/media/'.

```
STATIC_URL = 'static/'

DEFAULT_AUTO_FIELD = 'django.db.models.BigAutoField'

# Media settings
import os

BASE_DIR = Path(__file__).resolve().parent.parent

MEDIA_ROOT = os.path.join(BASE_DIR, 'user_uploads')
MEDIA_URL = '/media/'
```

52. Create a Django model called InfluencerProfile with fields: user (OneToOneField), display_name (CharField), bio (TextField), and profile_pic (ImageField). Ensure profile_pic uploads to 'profile_pics/'.

Ans=

```
from django.db import models
from django.contrib.auth.models import User

class InfluencerProfile(models.Model):
    user = models.OneToOneField(User, on_delete=models.CASCADE)
    display_name = models.CharField(max_length=100)
    bio = models.TextField(blank=True)
    profile_pic = models.ImageField(
        upload_to='profile_pics/',
        blank=True,
        null=True
    )

    def __str__(self):
        return self.display_name
```

53. Build a profile edit page where users can update their display name, bio, and upload a new profile picture, similar to editing your Instagram profile.

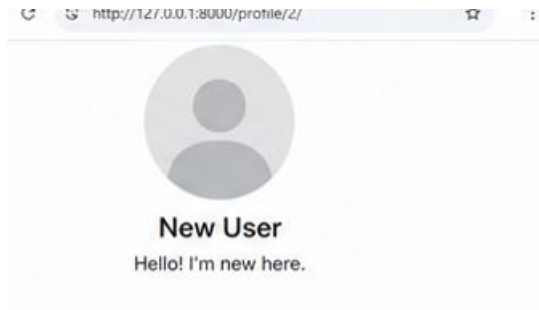


The screenshot shows a web form titled "Edit Your Profile". It contains three main sections: "Display Name" with a text input field containing "Techie Rohan"; "Bio" with a text area containing "Tech content creator | YouTuber | Blogger"; and "Profile Picture" with a "Choose file" button and the filename "new_pic.jpg" displayed. At the bottom of the form is a blue "Save Changes" button.

54. Display the uploaded profile picture on the user's profile page, falling back to a default image if no picture is uploaded.

Hint: Use the MEDIA_URL setting and check if profile_pic exists for the user.

Ans=



55. Add a phone number field to your InfluencerProfile model, update the edit form to allow changing it, and validate that the phone number is exactly 10 digits.

Constraint: Show an error message if the phone number is invalid.

Ans=

A screenshot of a web form for editing a user profile. The form contains several fields: 'Display Name' with the value 'Techie Rohan'; 'Bio' with the value 'Tech content creator | YouTuber | Blogger'; 'Phone Number' with the value '98765', which is highlighted with a red border and a red error message below it stating '* Phone number must be exactly 10 digits.'; and 'Profile Picture' with a 'Choose file' button and the text 'No file chosen'. At the bottom of the form is a blue 'Save Changes' button.

56. Create a Product model for a Flipkart-style app with fields: name, price, and category, then use Django's ModelForm to build a form for adding new products.

Ans=


```
class ProductForm(forms.ModelForm):
    class Meta:
        model = Product
        fields = ['name', 'price', 'category']
        widgets = {
            'name': forms.TextInput(attrs={
                'class': 'form-control'
            }),
            'price': forms.NumberInput(attrs={
                'class': 'form-control',
                'step': '0.01'
            }),
            'category': forms.TextInput(attrs={
                'class': 'form-control'
            })
        }
}
```

57. Build a view in Django that displays a list of all products from your Product model, showing name and price in a table, with an 'Edit' and 'Delete' button for each product.

1	iPhone 14	69999.00	Mobiles	Edit	Delete
2	Boat Rockerz 450	1499.00	Electronics	Edit	Delete
3	Campus Shoes	1799.00	Footwear	Edit	Delete
4	HP Laptop 15s	43990.00	Laptops	Edit	Delete

58.

Implement the edit feature for your Product model: clicking 'Edit' should open a pre-filled ModelForm allowing the user to update the product details and save changes.

Edit Product

Name

Price (₹)

Category

[Update Product](#) [Cancel](#)

59. Add a delete confirmation page: when the user clicks 'Delete' on a product, show a confirmation page with the product name and 'Yes, delete' and 'Cancel' options before actually deleting.'

Ans=

Delete Product

Are you sure you want to delete the product:

60. Create a simple HTML page with a list of songs (like a Spotify playlist) and add a 'Delete' button next to each song; use fetch() to send an AJAX DELETE request to a dummy endpoint when a button is clicked.



My Playlist

#		SONG	ARTIST	DURATION	ACTION
1		Shape of You	Ed Sheeran	3:53	Delete
2		Blinding Lights	The Weeknd	3:20	Delete
3		Someone You Loved	Lewis Capaldi	3:02	Delete
4		Sunflower	Post Malone	2:38	Delete

61.

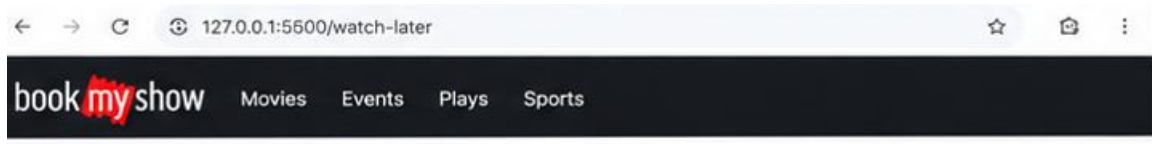
Build a mini Flipkart-style product list using Python (Flask or Django) and JavaScript, where clicking the 'Remove from Wishlist' button sends an AJAX request to delete a product and removes it from the UI without reloading the page.

Hint: Use fetch() with method 'DELETE' and update the DOM on success.

	iPhone 14 (128GB)	₹69,999	Remove from Wishlist
	Nike Air Zoom Pegasus	₹7,495	Remove from Wishlist
	boAt Rockerz 450	₹1,499	Remove from Wishlist
	HP Pavilion 15	₹59,990	Remove from Wishlist

62. Implement a feature for a BookMyShow-like movie booking page where deleting a movie from your 'Watch Later' list uses AJAX (fetch API) to send a JSON request and displays a dynamic success message when the deletion is complete.

Ans=



63. Debug this scenario: You have a function that sends a fetch() DELETE request to remove a playlist from a music app, but the playlist is not disappearing from the page after deletion. Identify and fix the issue so the UI updates correctly.

Hint: Check if your code is removing the DOM element after a successful response.

Ans=



63. Create a Django model called Restaurant with fields name, cuisine, and rating, then use filter() to get all restaurants with cuisine='Chinese' and rating greater than 4.

Ans=

```
class Restaurant(models.Model):
    name = models.CharField(max_length=100)
    cuisine = models.CharField(max_length=50)
    rating = models.FloatField()

    def __str__(self):
        return f"{self.name} ({self.cuisine})"
```

64.

In your Django shell, use exclude() to get all users from the User model who do NOT have 'gmail.com' in their email address.

Hint: Use the __contains lookup with exclude().

Ans=

Username	Email
john_doe	john@example.com
alice	alice@yahoo.com
bob_smith	bob@outlook.com
testuser	testuser@gmail.com

65. Build a Django QuerySet using `annotate()` to count the number of reviews for each Movie in a Movie and Review model setup, and print each movie's name with its review count.

```
class Movie(models.Model):
    title = models.CharField(max_length=150)
    release_year = models.IntegerField()

    def __str__(self):
        return self.title

class Review(models.Model):
    movie = models.ForeignKey(Movie, on_delete=models.CASCADE,
                             related_name='reviews')
    reviewer_name = models.CharField(max_length=100)
    rating = models.IntegerField()
    comment = models.TextField()
```

66. Given a Product and Category model (where Product has a foreign key to Category), use `select_related()` to fetch all products with their categories in a single query and display each product's name along with its category name.

Ans=

```
class Category(models.Model):
    name = models.CharField(max_length=100)

    def __str__(self):
        return self.name

class Product(models.Model):
    name = models.CharField(max_length=150)
    price = models.DecimalField(max_digits=10, decimal_places=2)
    category = models.ForeignKey(Category, on_delete=models.CASCADE)

    def __str__(self):
        return self.name
```

67. Use `Q()` objects to filter all Flipkart-style products that are either in the 'Electronics' category or have a price less than 1000, and paginate the results to show 5 products per page using Django's Paginator.

Ans=

```
from django.db.models import Q
from django.core.paginator import Paginator
from shop.models import Product

def product_list(request):
    query = Q(category__name='Electronics') | Q(price__lt=1000)
    products = Product.objects.filter(query).order_by('-id')

    paginator = Paginator(products, 5) # 5 per page
    page_number = request.GET.get('page')
    page_obj = paginator.get_page(page_number)
    return render(request, 'shop/product_list.html', {'page_obj': page_obj})
```

68. Create a Django view for a 'My Orders' page (like Zomato) and use the `@login_required` decorator to ensure only logged-in users can access it.

My Orders

Order ID	Restaurant	Items	Amount	Status	Ordered At
#1024	Pizza Palace	2 x Margherita Pizza	₹598	Delivered	18 May 2024
#1023	Burger Hub	1 x Veg Burger	₹199	Delivered	17 May 2024

69. Implement two custom user roles in your Django project: 'Seller' and 'Buyer' (inspired by Flipkart). Restrict access to a 'Post Product' page so only users with the 'Seller' role can access it, using Django's permissions system.

Post a New Product

Product Name
Wireless Headphones

Price (₹)
1999

Category
Electronics

Description
High quality wireless headphones.

Post Product

Product posted successfully!

70. Create two groups in the Django admin panel: 'MovieCritic' and 'MovieFan' (think BookMyShow). Assign different permissions to each group: 'MovieCritic' can add and edit movie reviews, while 'MovieFan' can only view reviews.

Home » Authentication and Authorization » Groups

AUTHENTICATION AND AUTHORIZATION

Groups [+ Add](#)

Users [+ Add](#)

MOVIES

Movies [+ Add](#)

Reviews [+ Add](#)

Select group to change

Action: Go 0 of 4 selected

☐	Name ?	Permissions	Available permissions ⓘ	Chosen permissions ⓘ
☐	MovieCritic	- Can add review - Can change review		✓ Can add review ✓ Can change review
☐	MovieFan	Can view review		✓ Can view review

70. Build a role-based dashboard view: when a user logs in, redirect them to a different dashboard based on their group ('Seller' sees a product management dashboard, 'Buyer' sees an order history dashboard).

Hint: Use Django's group membership to determine which dashboard to show after login.

Ans=

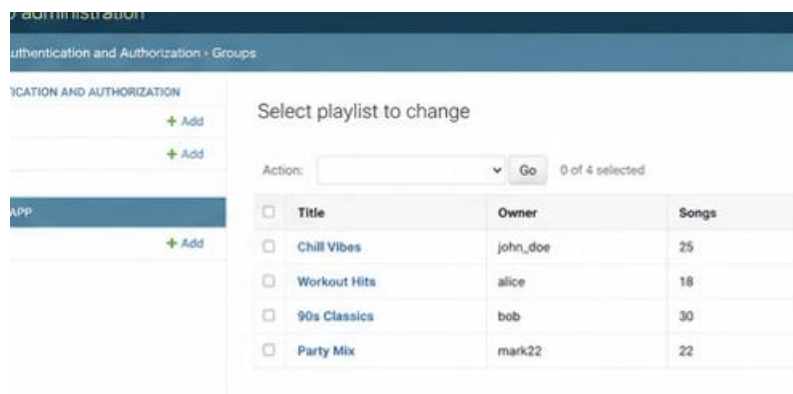
```

from django.contrib.auth.decorators import login_required
from django.shortcuts import redirect, render

@login_required
def role_based_dashboard(request):
    if request.user.groups.filter(name='Seller').exists():
        return redirect('products:seller_dashboard')
    elif request.user.groups.filter(name='Buyer').exists():
        return redirect('orders:buyer_dashboard')
    else:
        return render(request, 'dashboard/unknown.html')
def seller_dashboard(request):
    return render(request, 'dashboard/seller_dashboard.html')
def buyer_dashboard(request):
    return render(request, 'dashboard/buyer_dashboard.html')

```

71. Restrict access to the Django admin page for a custom model (like 'Playlist' for a Spotify-style app) so that only users in the 'Admin' group can view or edit it, while others get a permission denied message.



72. Integrate Google Login into a Django project so users can sign in using their Google account, and display their name and email on the homepage after login.

Hint: Use the 'social-auth-app-django' package and configure Google OAuth credentials in your settings.py.

```

INSTALLED_APPS = [
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
    # allauth
    'allauth',
    'allauth.account',
    'allauth.socialaccount',
    'allauth.socialaccount.providers.google',
]

AUTHENTICATION_BACKENDS = (
    'django.contrib.auth.backends.ModelBackend',
    'allauth.account.auth_backends.AuthenticationBackend',
)

# Google OAuth Credentials
SOCIALACCOUNT_PROVIDERS = {
    'google': {
        'APP': {
            'client_id': 'YOUR_GOOGLE_CLIENT_ID',
            'secret': 'YOUR_GOOGLE_CLIENT_SECRET',
            'key': '',
        }
    }
}

```

73.

Set up an email sending feature in Django using the Mailgun API — when a user registers, send them a welcome email with their username in the message body.

Hint: Use the 'requests' library to call Mailgun's API endpoint from your Django view.

Ans=

```
def register(request):
    if request.method == 'POST':
        form = RegisterForm(request.POST)
        if form.is_valid():
            user = form.save()
            send_welcome_email(user)
            messages.success(request, 'Registration successful! Please check your email.')
            return redirect('login')
        else:
            form = RegisterForm()
            return render(request, 'register.html', {'form': form})

def send_welcome_email(user):
    url = f'https://api.mailgun.net/v3/{settings.MAILGUN_DOMAIN}/messages'
    auth = ('api', settings.MAILGUN_API_KEY)
    data = {
        'from': f'MySite <mailgun@{settings.MAILGUN_DOMAIN}>',
        'to': [user.email],
        'subject': 'Welcome to MySite',
        'text': f'Hi {user.username},\n\nWelcome to MySite! We are glad to have you.'
    }
```

74. Build a Django view that sends a 6-digit OTP to a user's mobile number using the Twilio SMS API and displays a 'OTP sent' confirmation message on the web page.

```
def send_otp(request):
    if request.method == 'POST':
        mobile = request.POST.get('mobile')
        if not mobile:
            return render(request, 'send_otp.html', {'error': 'Enter mobile number'})

        otp = str(random.randint(100000, 999999))
        # Store OTP in session for demo (use DB/Cache in production)
        request.session['otp'] = otp

        client = Client(settings.TWILIO_ACCOUNT_SID, settings.TWILIO_AUTH_TOKEN)
        message = client.messages.create(
            body=f'Your OTP is {otp}. Valid for 5 minutes.',
            from_=settings.TWILIO_PHONE_NUMBER,
            to=mobile
        )
        return render(request, 'send_otp.html', {'success': True, 'mobile': mobile})
    return render(request, 'send_otp.html')
```

75. Use ChatGPT or Copilot to generate a Django model and view for storing and verifying OTP codes sent via SMS, then review the code and implement it in your project.

Constraint: Only use the AI tool for the initial code draft — you must understand and adapt the code before using it.

Send OTP

Mobile Number

+919876543210

Send OTP

Verify OTP

Mobile Number

OTP

[Verify OTP](#)

Verify OTP

Mobile Number

OTP

[Verify OTP](#)

❌ Invalid OTP

76. Set up a basic Flask app and create a /pay route that displays a simple payment form for ordering IPL match tickets, capturing name, email, and amount.

Ans=

→ http://127.0.0.1:5000/pay

IPL Ticket Payment

Enter your details to pay for IPL match tickets

Name

Email

Amount (INR)

[Pay Now](#)

77. Integrate the Paytm payment gateway sandbox in your Flask app so that when the user submits the ticket payment form, they are redirected to the Paytm test payment page and can complete a mock transaction.

Hint: Use the official Paytm Python SDK or API documentation for test credentials and callback

IPL Ticket Payment

Order ID: 9f3e0b2e-7c2a-4d6a-b2f2-1b3e2a9c8d11

Amount: ₹1500.00

Select an option to pay

☐ Paytm Wallet

☐ UPI

☐ Debit/Credit Card

☐ Net Banking

[Proceed](#)

78. After payment, handle the transaction callback from Paytm by creating a /payment-callback route that verifies the transaction status and displays a success or failure message to the user.

Ans=



79. Simulate a 'Pay Now' experience for a Zomato-style food order: modify your Flask app to accept a food order (dish name, price), initiate a Stripe test payment, and display the payment result on a new page.

Hint: Use Stripe's test API keys and make sure to handle both success and failure cases.



80. Use ChatGPT to generate Python code for integrating PayPal payment processing in a Flask route, then adapt the generated code to fit your IPL ticket payment scenario and test it with PayPal's sandbox environment.

Hint: Paste your ChatGPT prompt and the code you received as a comment in your Python file.


```

import requests
app = Flask(__name__)

CLIENT_ID = 'YourClientId'
CLIENT_SECRET = 'YourSecret'
BASE = 'https://api-m.sandbox.paypal.com'

def get_access_token():
    auth = (CLIENT_ID, CLIENT_SECRET)
    data = {'grant_type': 'client_credentials'}
    r = requests.post(f'{BASE}/v1/oauth2/token',
                     auth=auth, data=data)
    return r.json()['access_token']

@app.route('/paypal-pay')
def paypal_pay():
    token = get_access_token()
    headers = {'Content-Type': 'application/json',
              'Authorization': f'Bearer {token}'}

    data = {
        'intent': 'CAPTURE',
        'purchase_units': [{
            'amount': {'currency_code': 'INR', 'value': '1200.00'},
            'description': 'IPL Match Ticket'
        }],
        'application_context': {
            'return_url': url_for('paypal_success', _external=True),
            'cancel_url': url_for('paypal_cancel', _external=True)
        }
    }

    r = requests.post(f'{BASE}/v2/checkout/orders',
                     json=data, headers=headers)

```

81. Register for a free Google Maps API key and write a Python script that takes an address (e.g., 'IIM Ahmedabad, Gujarat') and prints its latitude and longitude using the Google Geocoding API.

Ans=

```

import requests

API_KEY = 'AIzaSy0-EXAMPLE-KEY' # Replace with your API key

address = 'IIM Ahmedabad, Gujarat'

url = 'https://maps.googleapis.com/maps/api/geocode/json'
params = {'address': address, 'key': API_KEY}

response = requests.get(url, params=params)
data = response.json()

if data['status'] == 'OK':
    location = data['results'][0]['geometry']['location']
    print(f"Address: {address}")
    print(f"Latitude : {location['lat']}")
    print(f"Longitude: {location['lng']}")
else:
    print('Geocoding failed:', data['status'])

```

82. Create a Django view called show_restaurant_location that takes a restaurant's address as input and displays a web page with an embedded Google Map showing the restaurant's location.

Hint: Use the Google Maps Embed API and pass the latitude and longitude to the template.

Ans=

```

from django.shortcuts import render
import requests

API_KEY = 'AIzaSyD-EXAMPLE-KEY'

def show_restaurant_location(request):
    address = request.GET.get('address', 'IIM Ahmedabad, Gujarat'),

    url = 'https://maps.googleapis.com/maps/api/geocode/json'
    params = {'address': address, 'key': API_KEY}

    r = requests.get(url, params=params).json()

    if r['status'] == 'OK':
        loc = r['results'][0]['geometry']['location']
        lat, lng = loc['lat'], loc['lng']
    else:
        lat, lng = None, None

    context = {'address': address, 'lat': lat, 'lng': lng}

    return render(request, 'restaurant/location.html', context)

```

83.

Write a Python function `find_nearby_cafes(user_lat, user_lng, cafes)` where `cafes` is a list of dictionaries with 'name', 'lat', and 'lng'. Return all cafes within 3 km of the user's location using the Haversine formula.
Constraint: Do not use any geolocation libraries—implement the distance calculation manually.

```

def haversine(lat1, lon1, lat2, lon2):
    R = 6371 # Earth radius in km
    dlat = math.radians(lat2 - lat1)
    dlon = math.radians(lon2 - lon1)
    a = (math.sin(dlat/2) ** 2 +
         math.cos(math.radians(lat1)) *
         math.cos(math.radians(lat2)) *
         math.sin(dlon/2) ** 2)
    c = 2 * math.atan2(math.sqrt(a), math.sqrt(1-a))
    return R * c

def find_nearby_cafes(user_lat, user_lng, cafes):
    nearby = []
    for cafe in cafes:
        dist = haversine(user_lat, user_lng, cafe['lat'], cafe['lng'])
        if dist <= 3:
            nearby.append((cafe, 'distance_km': round(dist, 2)))
    return sorted(nearby, key=lambda x: x['distance_km'])

user_lat, user_lng = 23.0225, 72.5714 # Example user location
cafes = [
    {'name': 'Cafe Coffee Day', 'lat': 23.0240, 'lng': 72.5750},
    {'name': 'Starbucks', 'lat': 23.0300, 'lng': 72.5800},
    {'name': 'The Brew Room', 'lat': 23.0500, 'lng': 72.6000},
    {'name': 'Mocha', 'lat': 23.0800, 'lng': 72.5700},
    {'name': 'Cafe Goodluck', 'lat': 22.9900, 'lng': 72.5600},
]

result = find_nearby_cafes(user_lat, user_lng, cafes)
for r in result:
    print(f'{r["name"]} - {r["distance_km"]} km')

```

84. Build a Django view called `search_by_distance` that allows users to enter their current address and see a list of Flipkart pickup points (use at least 5 hardcoded addresses) sorted by nearest distance on the results page.

Ans=

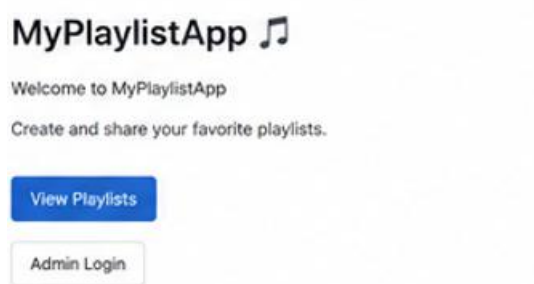


85. Create a new GitHub repository named 'MyPlaylistApp', initialize a local Django project (not Doctor Finder), and push the full project code to the new repository.

```
$ python manage.py runserver

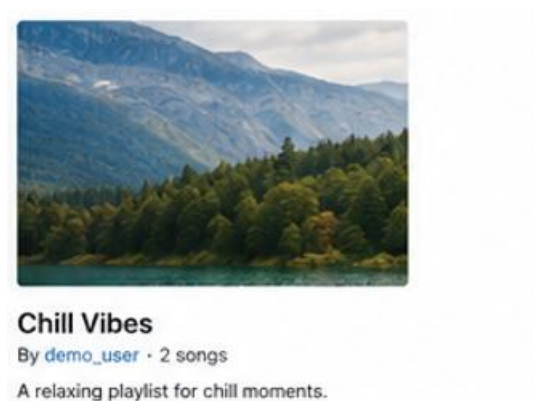
$ git init
$ git add .
$ git commit -m "Initial commit: MyPlaylistApp"
$ git branch -M main
$ git remote add origin https://github.com/<your-username>/MyPlaylistApp.git
$ git push -u origin main
Enumerating objects: 132, done.
Counting objects: 100% (132/132), done.
Delta compression using up to 8 threads
Compressing objects: 100% (110/110), done.
```

86. Deploy your Django 'MyPlaylistApp' project to PythonAnywhere, set up the web app, and ensure the home page is accessible via the public URL.

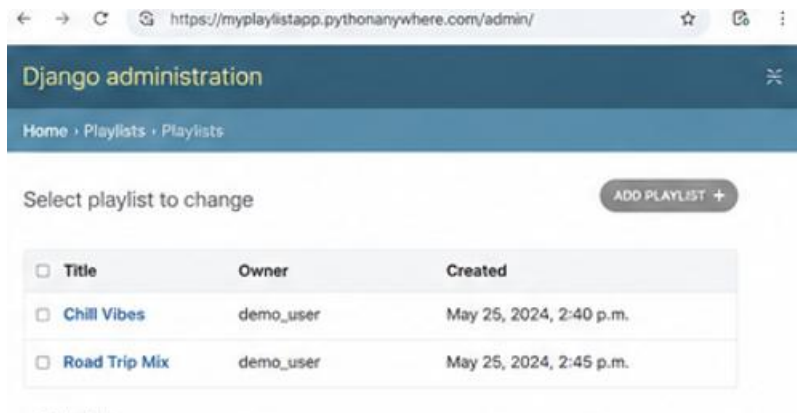


**87. Configure your Django project's static and media files on PythonAnywhere so that uploaded images (for example, playlist cover photos) display correctly on the live site.

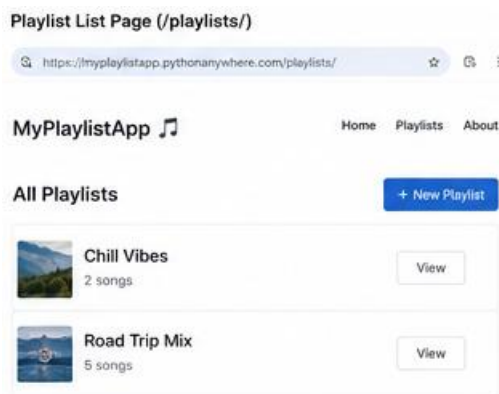
Hint: You may need to adjust STATIC_ROOT, MEDIA_ROOT, and update the PythonAnywhere web app settings for static and media mappings.**



88. Run Django's 'makemigrations' and 'migrate' commands on PythonAnywhere to create the database tables for your 'MyPlaylistApp' models, then verify that you can add a new playlist entry using the Django admin on the live site.

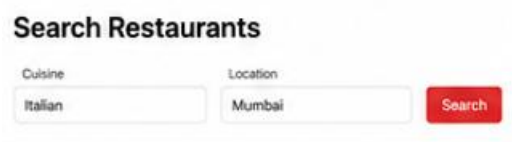


89. Test at least two live routes of your deployed 'MyPlaylistApp' (such as the playlist list page and the playlist detail page) and document any errors or issues you encounter, along with steps you took to resolve them.



90. Add a feature to your project that allows users to search for restaurants by cuisine and location, similar to Zomato's search — create a Restaurant model with name, cuisine, and location fields, and implement a search function that filters restaurants based on user input.

Ans=



91. Implement a login and OTP verification system for a movie ticket booking app — create registration and login forms, generate a 6-digit OTP on login, and verify the OTP before allowing access to the user

dashboard.

Hint: Store OTPs temporarily and expire them after 3 minutes.

Ans=

Register

Name

Email

Password

Register

Already have an account? [Login](#)

Login

Email

Password

Login

Don't have an account? [Register](#)

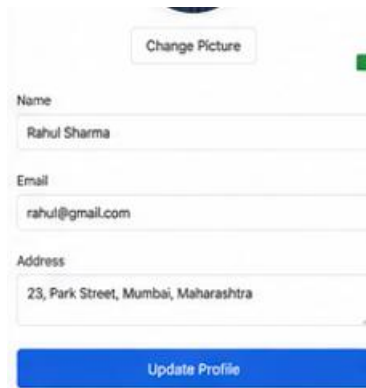
Verify OTP

Enter 6-digit OTP sent to
john@example.com

Verify OTP

Resend OTP (00:45)

92. Build a user profile management page for a Flipkart-style shopping app — allow users to update their name, email, delivery address, and profile picture, and display the updated details on the profile page.



A user profile update form with a 'Change Picture' button and a green profile picture placeholder. Below are input fields for Name (Rahul Sharma), Email (rahul@gmail.com), and Address (23, Park Street, Mumbai, Maharashtra). At the bottom is a blue 'Update Profile' button.

93. Integrate Google Maps into a cab booking app page — show the user's current location on the map and allow them to select a pickup location by clicking on the map.

Hint: Use the Google Maps JavaScript API and geolocation features.

```

script>
let map, marker;
function initMap(){
  navigator.geolocation.getCurrentPosition(function(pos){
    const userLoc = {
      lat: pos.coords.latitude,
      lng: pos.coords.longitude
    };
    map = new google.maps.Map(document.getElementById('map'), {
      zoom: 14,
      center: userLoc
    });
    marker = new google.maps.Marker({
      position: userLoc,
      map: map,
      draggable: true
    });
    map.addListener('click', function(e){
      marker.setPosition(e.latLng);
      document.getElementById('pickup').innerText =
        e.latLng.lat().toFixed(4) + ', ' + e.latLng.lng().t
        toFixed(4);
    });
  });
}

```

94. Deploy your completed project (not the Doctor Finder) to PythonAnywhere, push the final code to a new GitHub repository, and share both the live deployment URL and GitHub repo link.

